

EFREI Paris

Staffordshire University

Introduction to Machine Learning

Statistical Clustering Analysis of Pokémon Data Using K-Means, DBSCAN and Agglomerative Methods

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All code, notebooks, and datasets are available in the project GitHub repository:
[NathanaelLUCETTE/projet ML](#)

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Introduction

This project aims to analyze how Pokémon can be grouped into meaningful categories using unsupervised learning techniques. We focus on two complementary families of features: **Stats**, which describe combat performance, and **Power/Rarity**, which capture rarity, strength, and progression-related attributes.

To ensure that the clusters we obtain reflect real structures in the data rather than artifacts of a specific algorithm, we compare three clustering methods: **K-Means**, **DBSCAN**, and **Agglomerative Clustering**. K-Means provides clear and interpretable segmentations, while DBSCAN and Agglomerative serve as validation tools: if all three methods reveal similar structures, we can trust that the clusters correspond to genuine patterns in the dataset.

After verifying with DBSCAN and Agglomerative that the clusters found by K-Means correspond to real structures in the data, we use the **K-Means centroids** to assign new Pokémon to these clusters. This generalization step is performed on two external datasets: a **fan-made dataset** and the **official Generation 9 Pokémon**. This allows us to evaluate how well the learned clusters transfer to unseen data and to identify Pokémon whose profiles deviate significantly from the established patterns.

PHASE 1 — Data Exploration and Preprocessing

1. Dataset Loading and Initial Structure

The dataset was first imported and inspected to understand its global structure. Basic information such as the number of rows, columns, data types, and missing values was examined to ensure that the dataset was suitable for clustering analysis.

pokedex_id	name	genus	generation	type_1	type_2	num_types	hp	attack	defense	...	sprite_url	is_dual_type	bmi	attack_defense_ratio	physical_total	special_total	offensive
0	1	bulbasaur	Seed Pokémon	I	grass	poison	2	45	49	49	...	https://raw.githubusercontent.com/PokeAPI/spri...	True	14.1	1.00	98	130
1	2	ivysaur	Seed Pokémon	I	grass	poison	2	60	62	63	...	https://raw.githubusercontent.com/PokeAPI/spri...	True	13.0	0.98	125	160
2	3	venusaur	Seed Pokémon	I	grass	poison	2	80	82	83	...	https://raw.githubusercontent.com/PokeAPI/spri...	True	25.0	0.99	165	200
3	4	charmander	Lizard Pokémon	I	fire	NaN	1	39	52	43	...	https://raw.githubusercontent.com/PokeAPI/spri...	False	23.6	1.21	95	110
4	5	charmeleon	Flame Pokémon	I	fire	NaN	1	58	64	58	...	https://raw.githubusercontent.com/PokeAPI/spri...	False	15.7	1.10	122	145

5 rows x 43 columns

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 43 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   pokedex_id                            1025 non-null   int64
1   name                                  1025 non-null   object
2   genus                                 1025 non-null   object
3   generation                            1025 non-null   object
4   type_1                                1025 non-null   object
5   type_2                                526 non-null    object
6   num_types                             1025 non-null   int64
7   hp                                    1025 non-null   int64
8   attack                               1025 non-null   int64
9   defense                              1025 non-null   int64
10  sp_attack                             1025 non-null   int64
11  sp_defense                            1025 non-null   int64
12  speed                                 1025 non-null   int64
13  base_stat_total                       1025 non-null   int64
14  height_m                             1025 non-null   float64
15  weight_kg                             1025 non-null   float64
16  base_experience                       1025 non-null   int64
17  ability_1                             1025 non-null   object
18  ability_2                             538 non-null    object
19  hidden_ability                       856 non-null    object
20  color                                 1025 non-null   object
21  shape                                 1025 non-null   object
22  habitat                               386 non-null    object
23  growth_rate                           1025 non-null   object
24  egg_groups                            1025 non-null   object
25  is_legendary                          1025 non-null   bool
26  is_mythical                           1025 non-null   bool
27  is_baby                               1025 non-null   bool
28  capture_rate                          1025 non-null   int64
29  base_happiness                        1025 non-null   int64
30  hatch_counter                         1025 non-null   int64
31  gender_rate                           1025 non-null   int64
32  description                            1025 non-null   object
33  sprite_url                            1025 non-null   object
34  is_dual_type                          1025 non-null   bool
35  bmi                                    1025 non-null   float64
36  attack_defense_ratio                  1025 non-null   float64
37  physical_total                        1025 non-null   int64
38  special_total                         1025 non-null   int64
39  offensive_total                       1025 non-null   int64
40  defensive_total                       1025 non-null   int64
41  gender_distribution                   1025 non-null   object
42  stat_tier                             1025 non-null   object
dtypes: bool(4), float64(4), int64(18), object(17)
```

	pokedex_id	num_types	hp	attack	defense	sp_attack	sp_defense	speed	base_stat_total	height_m	...	capture_rate	base_happiness	hatch_counter	gender_rate	bmi	attack_defense_ratio	physical_total	special_total	offensive
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	...	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	513.000000	1.513171	70.184390	77.521951	72.587317	70.080976	70.205854	67.186341	427.686629	1.211610	...	95.454634	58.897561	29.487805	2.899512	138.378537	1.152361	150.028068	140.286629	140.286629
std	296.036315	0.500070	26.631054	29.782541	29.286972	29.658378	28.638329	28.717227	112.770735	1.248167	...	75.991841	22.482942	26.498674	2.231839	3122.211180	0.454197	50.585755	48.865640	48.865640
min	1.000000	1.000000	1.000000	5.000000	5.000000	10.000000	20.000000	5.000000	175.000000	0.100000	...	3.000000	0.000000	5.000000	-1.000000	0.000000	0.040000	10.000000	35.000000	35.000000
25%	257.000000	1.000000	50.000000	55.000000	50.000000	47.000000	50.000000	45.000000	325.000000	0.500000	...	45.000000	50.000000	20.000000	1.000000	19.100000	0.840000	110.000000	101.000000	101.000000
50%	513.000000	2.000000	68.000000	75.000000	70.000000	65.000000	67.000000	65.000000	450.000000	1.000000	...	60.000000	70.000000	20.000000	4.000000	29.200000	1.080000	150.000000	135.000000	135.000000
75%	769.000000	2.000000	85.000000	100.000000	90.000000	90.000000	86.000000	88.000000	508.000000	1.500000	...	140.000000	70.000000	25.000000	4.000000	44.400000	1.420000	185.000000	175.000000	175.000000
max	1025.000000	2.000000	255.000000	181.000000	230.000000	173.000000	230.000000	200.000000	720.000000	20.000000	...	255.000000	140.000000	120.000000	8.000000	99990.000000	4.500000	342.000000	300.000000	300.000000

8 rows x 22 columns

This initial inspection confirmed that the dataset contains a rich set of numerical attributes describing Pokémon characteristics across multiple generations.

2. Exploratory Data Analysis (EDA)

A preliminary EDA was conducted to identify global patterns, detect anomalies, and understand the distribution of key variables. This step is essential to determine whether the dataset contains meaningful structures that clustering algorithms can exploit.

Irrelevant or non-numerical columns (e.g., names, types, habitat) were removed to focus exclusively on quantitative attributes suitable for clustering.

The remaining numerical features were then reviewed to ensure consistency and completeness. Missing values were handled appropriately (either removed or imputed depending on the column).

This produced a clean numerical dataset ready for feature grouping.

Key observations:

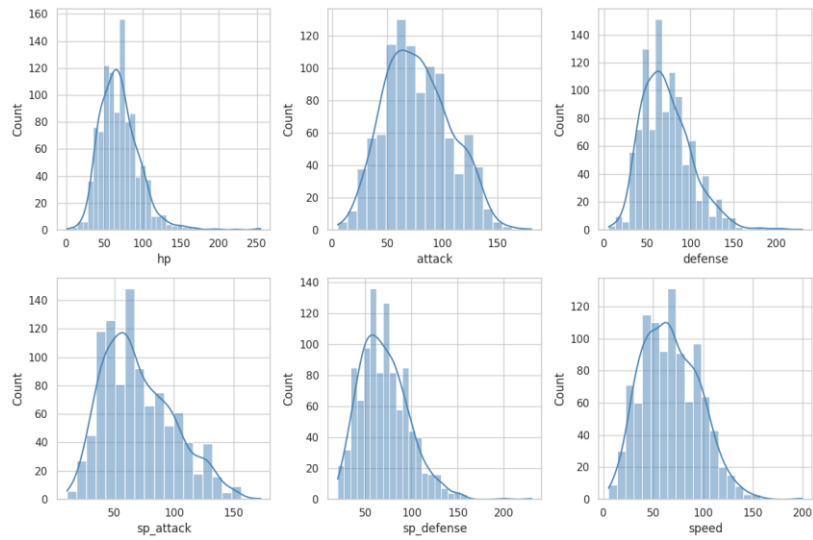
- Numerical features exhibit heterogeneous scales (e.g., HP vs. capture_rate), confirming the need for normalization.
- Some features show strong correlations (e.g., base_stat_total with base_experience).
- Distributions vary significantly between combat-related stats and rarity-related attributes.

3. Feature Group Definition

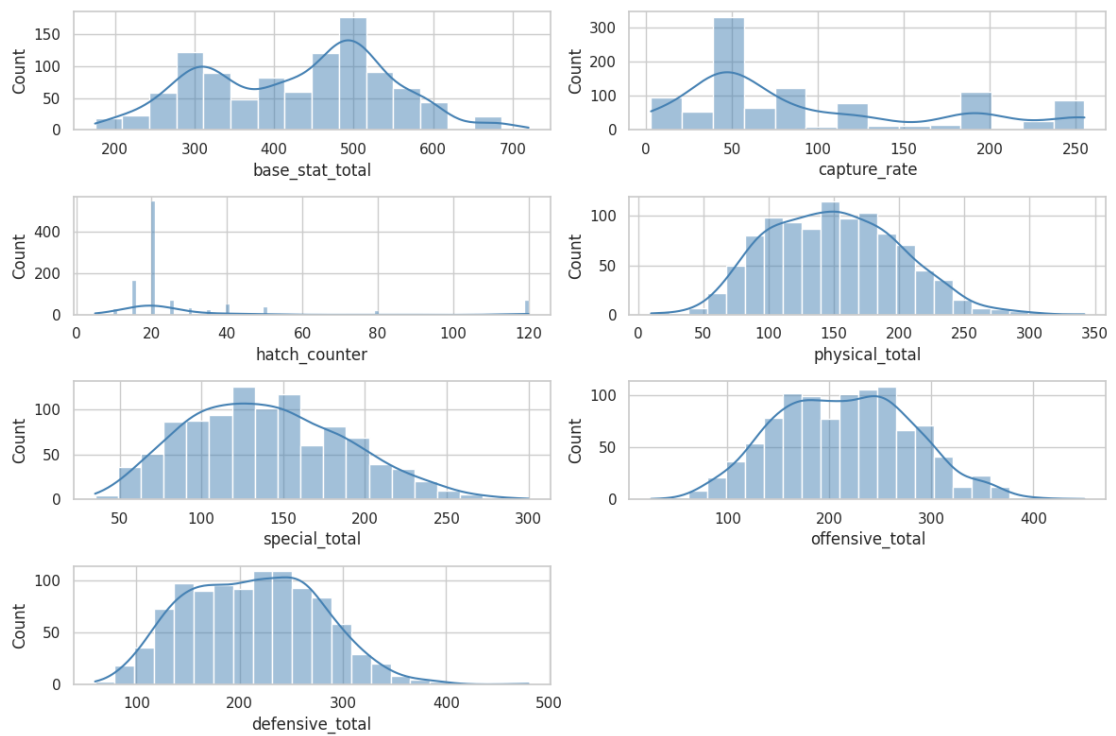
To capture different dimensions of Pokémon characteristics, the numerical features were divided into two meaningful families:

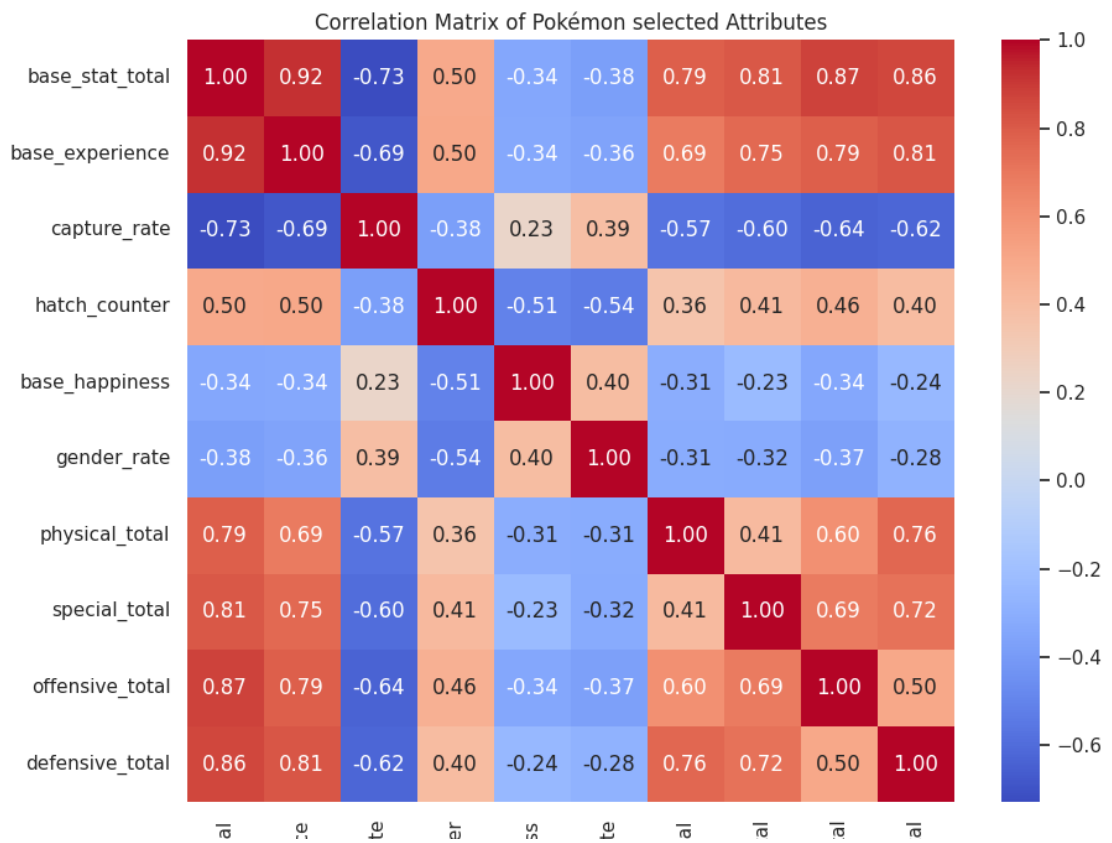
- Stats Features (Combat-related) which describe a Pokémon's battle performance (hp/attack/defense /special_attack /special_defense/speed)

These variables represent the classical combat stats used in competitive play.



- Power/Rarity features which reflect how rare, powerful, or difficult it is to obtain a Pokémon (base_stat_total, capture_rate, hatch_counter, physical_total, special_total, offensive_total, defensive_total).





PHASE 2 — Processing and Clusterability Analysis

Problem Type & Model Choice

The dataset does not contain predefined labels or categories that could be used for supervised learning. Instead, it consists exclusively of numerical attributes describing Pokémon characteristics. Since no ground-truth classes exist, the task is naturally formulated as a clustering problem. The objective is to uncover latent structures in the data and identify meaningful groups of Pokémon based on their statistical profiles.

To ensure a robust and comprehensive analysis, three complementary clustering algorithms were selected:

- **K-Means Clustering** Chosen for its interpretability and its ability to produce clear, centroid-based segmentations. K-Means is well-suited for datasets where clusters are expected to be compact and roughly spherical, making it a strong baseline for Pokémon stat distributions.
- **DBSCAN (Density-Based Spatial Clustering)** Used to detect density-based structures and identify outliers. DBSCAN does not require specifying the number of clusters and is effective for revealing irregular shapes or rare Pokémon profiles that deviate from typical stat patterns.
- **Agglomerative Hierarchical Clustering (Ward linkage)** Selected to validate the hierarchical relationships between Pokémon profiles. This method provides a complementary perspective by building clusters step-by-step and revealing how groups merge at different levels of granularity.

Using these three algorithms allows cross-validation of the results: if all methods reveal similar structures, we can be confident that the clusters reflect genuine patterns in the Pokémon dataset rather than artifacts of a single algorithm.

4. Scaling

Because the features operate on very different scales (e.g., HP ranges around 50–150, while capture_rate ranges 3–255), standardization was applied using **StandardScaler**.

Two separate scalers were trained:

- one for **Stats**,
- one for **Power/Rarity**.

This ensures that each feature contributes equally to the clustering algorithms.

5. Clusterability Evaluation

Before applying clustering algorithms, we evaluated whether the data naturally forms clusters. Three tools were used:

- **Elbow Method**
- **Silhouette Score**
- **k-distance plot (DBSCAN)**

These analyses were performed separately for Stats and Power/Rarity.

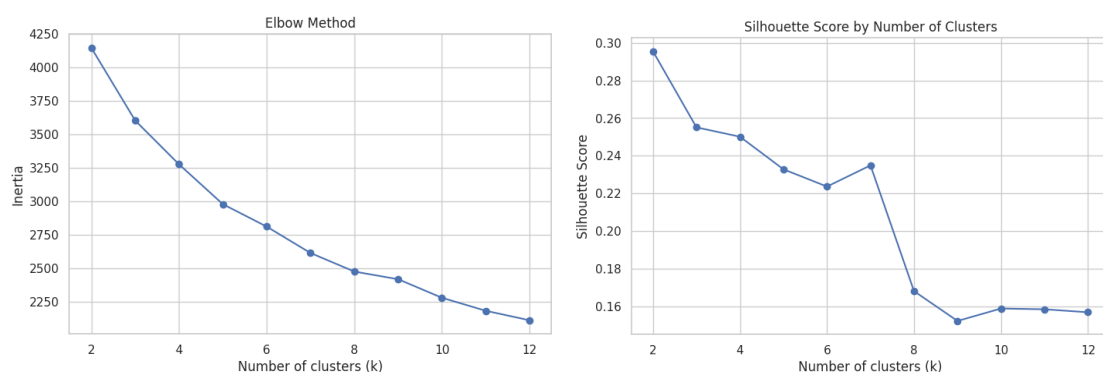
5.1 Stats — Clusterability Results

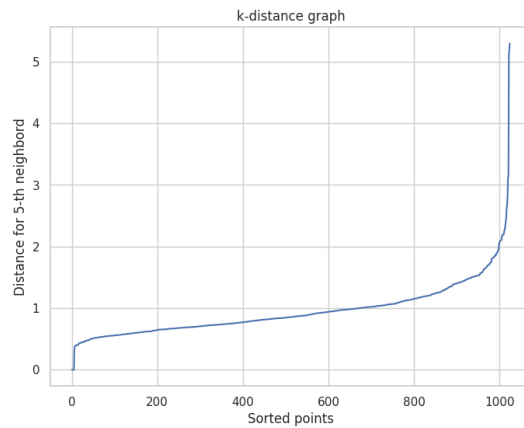
The Elbow Method does not reveal a clear inflection point for the Stats feature group. The inertia decreases smoothly and almost linearly, indicating that the data does not naturally form compact, well-separated clusters. This suggests that Stats follow a continuous structure rather than discrete groups.

The Silhouette Score reaches its maximum at $k = 2$, confirming that the strongest separation in the Stats features corresponds to two broad combat archetypes (e.g., fast attackers vs. defensive tanks). Beyond $k = 2$, the score decreases sharply, indicating that additional clusters are less well-defined.

The k-distance plot shows no clear elbow, and the curve increases smoothly without a sharp transition. This indicates that the Stats features do not contain density-based clusters.

However, we will also test with $k=3$ or $k=4$ because we want to have interesting categories, more than attackers Vs tanks.





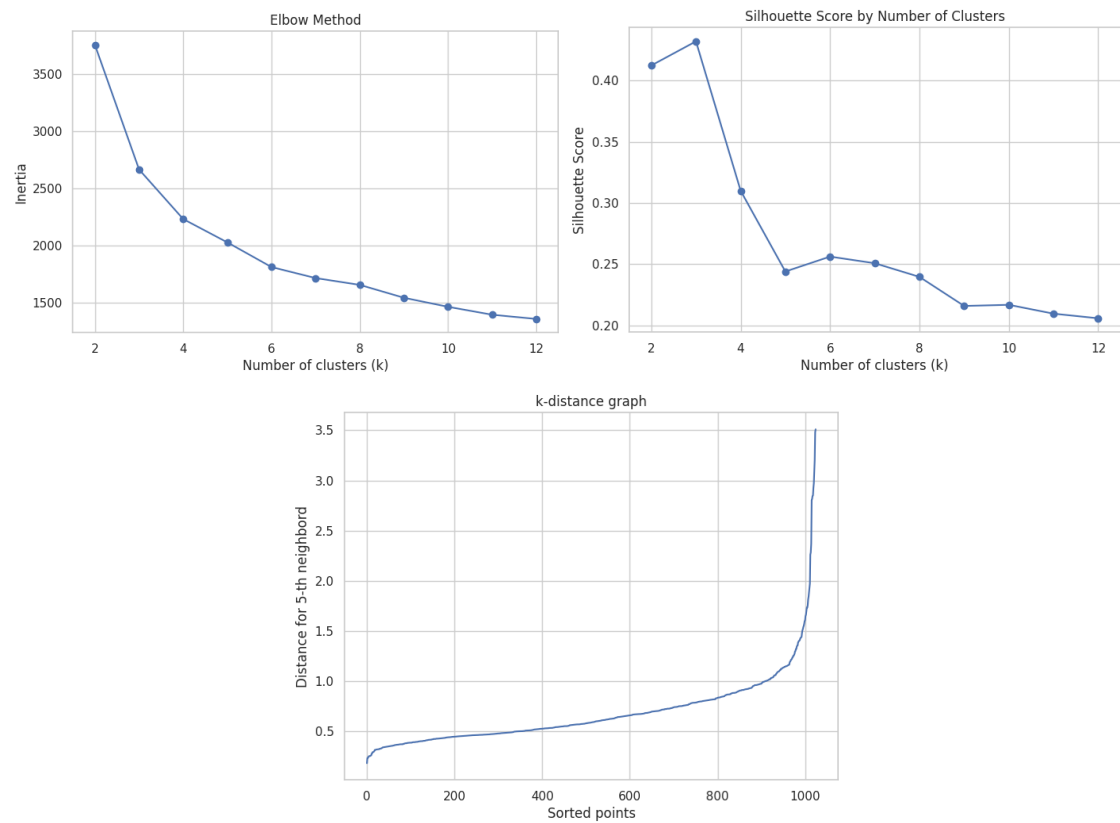
6.2 Power/Rarity — Clusterability Results

The Elbow Method shows a noticeable inflection around $k = 3$, where the decrease in inertia becomes significantly less steep. This suggests that the Power/Rarity features naturally form three compact and well-separated groups.

The Silhouette Score reaches its maximum at $k = 3$, confirming that three clusters provide the best balance between compactness and separation. This supports the interpretation of three natural categories: low-power common Pokémon, intermediate Pokémon, and high-power rare Pokémon.

The k-distance plot ($k = 5$) exhibits a clear elbow around $\text{eps} \approx 2.5$. This value represents the transition between dense regions and isolated points. Using $\text{eps} = 2.5$ and $\text{min_samples} = 5$ yields the most coherent DBSCAN configuration: one main cluster and a few outliers corresponding to extremely powerful or rare Pokémon.

However, we will test another value (notably $k=4$ and $\text{eps}=1.5$) to really understand the hidden clusters and the different interpretations of the dataset.



Overall, the clusterability analysis shows that Stats form a continuous structure with no strong density-based clusters, while Power/Rarity exhibits a clear three-cluster organization. These results justify exploring $k = 2-3$ for Stats and $k = 3-4$ for Power/Rarity, and selecting $\text{eps} \approx 2.5$ for DBSCAN validation.

PHASE 3 — Clustering and Interpretation

8. Clustering Methods

To ensure that the structures observed in the dataset are robust and not artifacts of a single algorithm, three complementary clustering methods were applied:

- **K-Means**, which partitions the data into k spherical clusters based on distance to centroids.
- **DBSCAN**, a density-based method capable of detecting outliers and non-spherical clusters.
- **Agglomerative Clustering**, a hierarchical method that groups points based on linkage criteria.

Each method was applied separately to the **Stats** and **Power/Rarity** feature groups.

8.1 K-Means Clustering

K-Means was run with several values of k to evaluate the stability and interpretability of the resulting clusters. PCA was used to visualize the clusters in two dimensions.

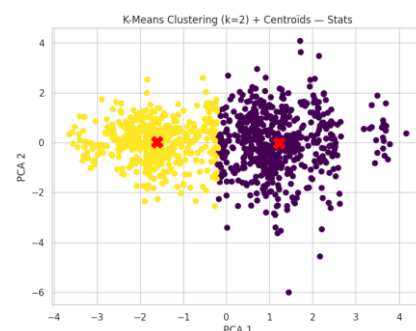
8.1.1 Stats — K-Means Results

K-Means was tested with $k = 2$, $k = 3$, and $k = 4$.

$k = 2$

The clustering is clean and well-separated. The two clusters correspond to broad combat archetypes:

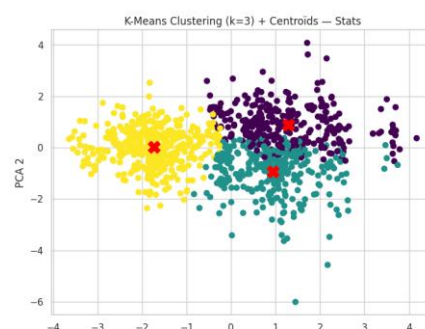
- fast, offensive Pokémon
- defensive, tank-like Pokémon



$k = 3$

The clusters remain well-separated. One of the clusters from $k = 2$ splits into two subgroups, revealing a more nuanced structure:

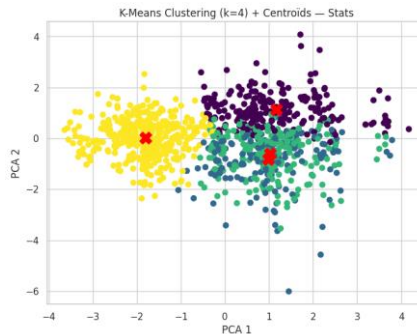
- fast special attackers
- physical attackers



- defensive tanks

k = 4

The clusters begin to overlap significantly. The separation becomes less meaningful, indicating that $k = 4$ over-segments the data.



Interpretation

K-Means reveals **clear combat archetypes**, especially at $k = 2$ and $k = 3$. $k = 3$ provides the most interpretable segmentation without overfitting.

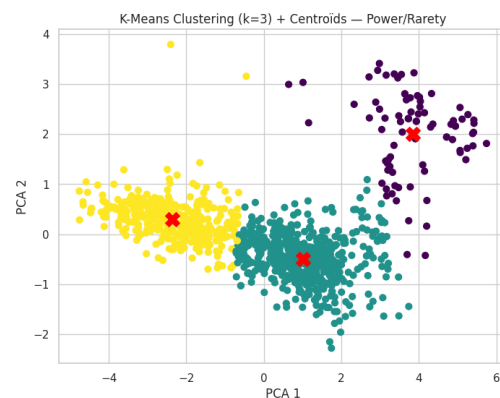
8.1.2 Power/Rarity — K-Means Results

K-Means was tested with $k = 3$ and $k = 4$, based on Phase 2 findings.

k = 3

The clusters are compact, well-separated, and form a triangular structure in PCA space. They correspond to:

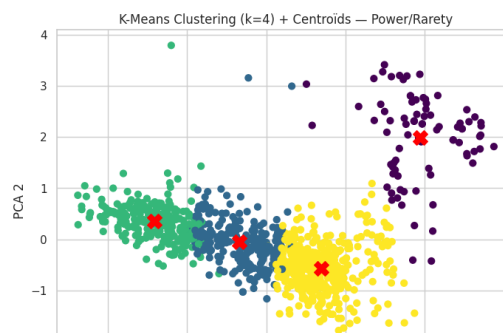
- low-power common Pokémon.
- intermediate Pokémon.
- high-power rare Pokémon.



k = 4

The intermediate group splits into two meaningful subgroups:

- weaker intermediate Pokémon.
- stronger intermediate Pokémon.



The rare/powerful cluster remains compact.

Interpretation

Although $k = 3$ is optimal mathematically, **$k = 4$ provides the most interpretable segmentation**, capturing a natural progression from common to rare/powerful Pokémon.

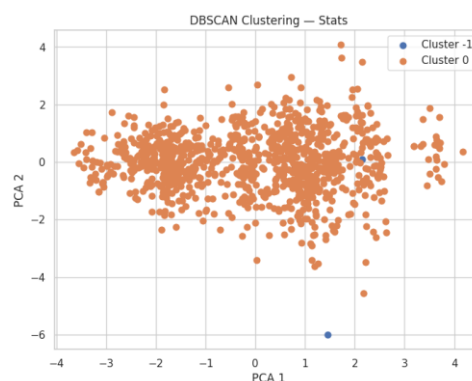
8.2 DBSCAN

DBSCAN was used to validate whether the data contains density-based clusters and to detect outliers.

8.2.1 Stats — DBSCAN Results

Using $\text{eps} = 2.5$ and $\text{min_samples} = 4$:

- DBSCAN identifies **one main cluster**
- plus, a few outliers corresponding to Pokémon with extreme stats (e.g., legendaries)



Interpretation

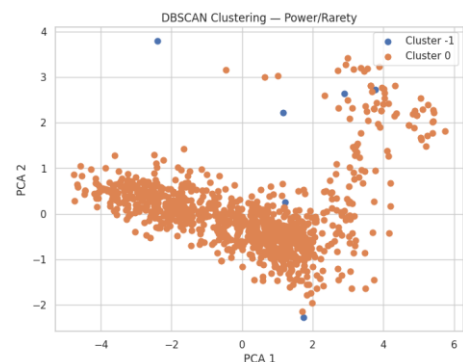
Stats form a **continuous distribution** with no natural density-based clusters. This confirms that K-Means imposes structure rather than revealing inherent density groups.

8.2.2 Power/Rarity — DBSCAN Results

Two configurations were tested:

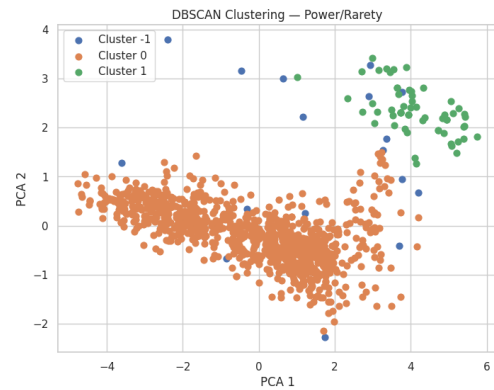
eps = 2.5

- One large cluster.
- A few outliers (very powerful or rare Pokémon) .



eps = 1.5

- A compact cluster of extremely powerful Pokémon.
- The rest grouped into a broad main cluster.
- A few outliers.



Interpretation

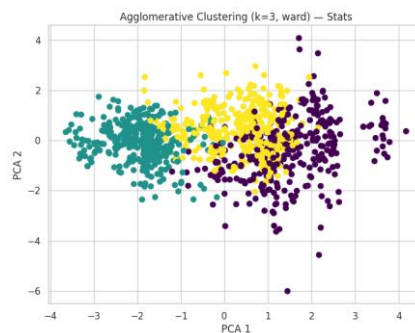
Power/Rarity features form a **continuous progression**, not discrete density clusters. DBSCAN is useful for detecting **extreme high-power Pokémon**, but not for segmenting the dataset into multiple interpretable groups.

8.3 Agglomerative Clustering

Agglomerative Clustering (Ward linkage) was only applied with $k = 3$ (the test with $k = 2$ and $k = 4$ are not interesting)

8.3.1 Stats — Agglomerative Results

Agglomerative clustering with $k = 3$ reveals a similar three-archetype segmentation, more interesting than with only two clusters (fast Vs slow pokemon).



Interpretation

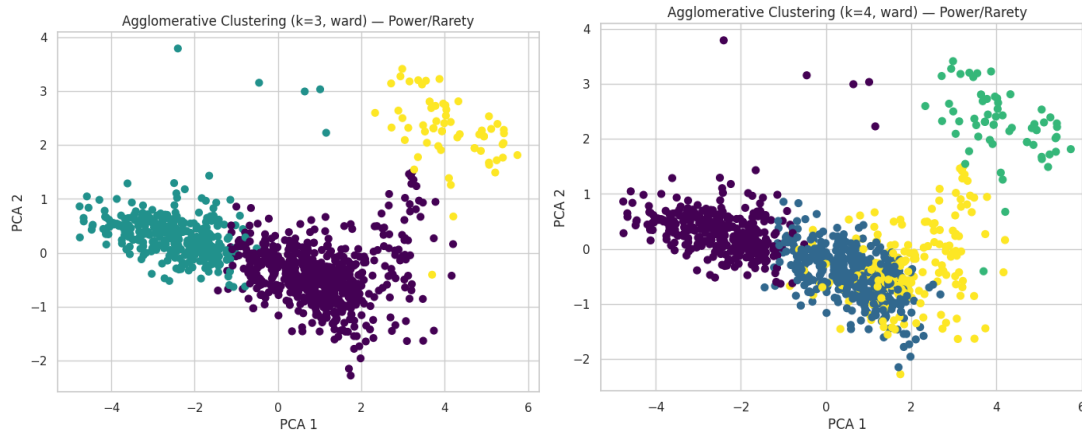
Agglomerative confirms the **stability** of the structures found by K-Means.

8.3.2 Power/Rarity — Agglomerative Results

Agglomerative with Ward linkage produces results highly consistent with K-Means:

- $k = 3$ recovers the three natural groups

- $k = 4$ splits the intermediate group into two meaningful sub-clusters



Interpretation

Agglomerative validates both $k = 3$ and $k = 4$ as coherent and interpretable structures.

9. Global Interpretation

Stats

- Reveal **clear combat archetypes**.
- Best represented by $k = 3$.
- DBSCAN confirms a continuous structure with a few extreme outliers.

Power/Rarity

- Form a **continuous progression** from common to rare/powerful.
- K-Means extracts **3–4 meaningful tiers**.
- DBSCAN isolates extreme high-power Pokémon.

Cross-method validation

- K-Means provides the clearest segmentation.
- Agglomerative confirms the robustness of the structures.
- DBSCAN highlights outliers and validates the continuous nature of the data.

Conclusion

Pokémon do not cluster along a single dimension. Instead, they form **complementary structures** depending on the feature family:

- **Stats** → **combat roles**.
- **Power/Rarity** → **rarity and power tiers**.

Phase 3.5 -- Association Between K-Means Clusters and Pokémon Categories

To interpret the clusters produced by K-Means, we analyzed the **centroids** of each cluster and ranked the features inside each centroid using the `rank_centroids()` function. These ranking highlights which attributes dominate each cluster and allow us to associate each group with a meaningful Pokémon category.

1. Stats Feature Group (k = 4)

The centroids obtained from K-Means (k = 4) reveal four distinct combat archetypes. Ranking the features inside each centroid shows clear patterns:

	hp	attack	defense	sp_attack	sp_defense	speed
0	0.182162	0.340839	-0.038310	0.962949	0.377031	1.214185
1	0.382205	-0.180728	0.823121	0.530108	1.126868	-0.474843
2	0.792816	1.106646	0.645343	-0.200507	0.071371	0.036363
3	-0.760067	-0.786482	-0.736869	-0.733129	-0.799275	-0.568755
	hp	attack	defense	sp_attack	sp_defense	speed
0	5.0	4.0	6.0	2.0	3.0	1.0
1	4.0	5.0	2.0	3.0	1.0	6.0
2	2.0	1.0	3.0	6.0	4.0	5.0
3	4.0	5.0	3.0	2.0	6.0	1.0

Cluster 0 — Fast Special Attackers

- High **speed** and **special attack**.
- Low **defense** → Corresponds to fast, fragile sweepers relying on special damage.

Cluster 1 — Tanks

- High **HP**, **defense**, and **special defense**.
- Low **speed** → Defensive Pokémon designed to absorb damage.

Cluster 2 — Physical Attackers

- High **attack** and **speed**.
- Lower special stats → Offensive Pokémon relying on physical moves.

Cluster 3 — Low-Stat Pokémon

- All stats ranked low → Early-game or unevolved Pokémon with weak overall profiles.

2. Power/Rarity Feature Group (k = 4)

The Power/Rarity centroids also reveal four coherent categories. Ranking the centroid values highlights a clear progression from common to rare/powerful Pokémon.

	base_stat_total	capture_rate	hatch_counter	physical_total	special_total	offensive_total	defensive_total
0	1.567649	-1.006581	3.073013	1.085499	1.341546	1.453212	1.261073
1	-0.373487	-0.192075	-0.270347	-0.360948	-0.319230	-0.284487	-0.364857
2	-1.273591	1.386929	-0.378890	-1.019740	-1.027016	-1.097820	-1.110909
3	0.702657	-0.546059	-0.204258	0.632430	0.562258	0.563580	0.656828
	base_stat_total	capture_rate	hatch_counter	physical_total	special_total	offensive_total	defensive_total
0	2.0	7.0	1.0	6.0	4.0	3.0	5.0
1	7.0	1.0	2.0	5.0	4.0	3.0	6.0
2	7.0	1.0	2.0	3.0	4.0	5.0	6.0
3	1.0	7.0	6.0	3.0	5.0	4.0	2.0

Cluster 0 — Very Powerful / Very Rare Pokémon

- Highest base_stat_total, offensive_total, defensive_total.
- Lowest capture_rate → Legendary-like Pokémon with exceptional stats.

Cluster 1 — Strong Non-Legendary Pokémon

- High stats but not extreme.
- Moderate capture rate → Final evolutions and high-tier non-legendary Pokémon.

Cluster 2 — Weak, Common Pokémon

- Low totals.
- High capture rate → Early-game or common species.

Cluster 3 — Mid-Tier Common Pokémon

- Moderate stats.
- Higher hatch/capture values → Middle evolutions or average Pokémon.

1. Insights

The clustering analysis reveals several meaningful patterns in the Pokémon dataset. First, the Stats feature group naturally organizes Pokémon into combat archetypes such as fast special attackers, physical sweepers, tanks, and low-stat early-game species. These clusters are consistent across K-Means, DBSCAN, and Agglomerative Clustering, confirming that the underlying structure of combat stats is continuous but still exhibits identifiable tendencies.

The Power/Rarity feature group shows even clearer separations, with well-defined clusters corresponding to common low-tier Pokémon, mid-tier species, strong non-legendary Pokémon, and legendary-like profiles. These clusters reflect game-design intentions: rarity, capture difficulty, and stat totals align strongly with perceived power levels.

The generalization tests demonstrate that the learned clusters are stable and transferable. Both fan-made Pokémon and official Generation 9 Pokémon fall into the same cluster structure as earlier generations. Outliers correspond to meaningful exceptions such as Paradox Pokémon or artificially modified fan-made species. This confirms that the clustering model captures robust structural patterns rather than dataset-specific noise.

2. Recommendations

Based on the clustering results, several practical recommendations can be made:

- **Game balance analysis:** The clusters can help identify Pokémon whose stats or rarity deviate significantly from their intended archetype. Designers could use this to adjust overly strong or weak species.
- **Creation of new Pokémon:** The centroid profiles provide templates for designing balanced Pokémon that fit existing archetypes. Fan-made creators or game designers can use these clusters as guidelines.
- **Detection of anomalies:** Outliers detected by DBSCAN or distance-to-centroid analysis can highlight Pokémon with unusual or inconsistent stat distributions, useful for quality control or dataset validation.
- **Feature engineering for future models:** The clear separation in Power/Rarity suggests that these features could be used to build predictive models for rarity tiers or encounter difficulty.

3. Limitations and Future Work

Despite strong results, several limitations must be acknowledged:

- **Feature scope:** The analysis focuses exclusively on numerical attributes. Important qualitative features such as types, abilities, or move pools were not included, which may limit the richness of the clusters.
- **Model assumptions:** K-Means assumes spherical clusters and equal variance, which may not fully reflect the true structure of Pokémon stats. Alternative models such as Gaussian Mixture Models or Spectral Clustering could capture more complex shapes.
- **Choice of k:** While k was guided by clusterability metrics, the selection still involves subjective interpretation. Different values of k may reveal alternative but equally valid structures.
- **Dataset bias:** The dataset includes Pokémon from multiple generations with different design philosophies. Some clusters may reflect generation-specific trends rather than universal patterns.

Future Work

Future analyses could explore:

- Incorporating categorical features (types, abilities) through embeddings or mixed-type clustering methods.
- Testing probabilistic clustering models to capture overlapping archetypes.
- Building predictive models to classify new Pokémon into archetypes based on partial information.
- Extending the analysis to competitive viability metrics or move-set diversity.
- Applying dimensionality reduction techniques beyond PCA (UMAP, t-SNE) for more nuanced visualizations.

PHASE 4 — Generalization / Test (K-Means)

11. Motivation for Generalization

After validating with DBSCAN and Agglomerative Clustering that the clusters produced by K-Means correspond to real structures in the data, the next step is to evaluate whether these clusters can be reliably applied to new Pokémon.

This generalization step answers a key question:

Do the clusters learned from Generations 1–8 remain valid when applied to new or external Pokémon?

To test this, the K-Means centroids learned earlier are used to assign cluster labels to:

1. A fan-made dataset (external, non-official Pokémon)
2. The official Generation 9 Pokémon (new real data)

This allows us to evaluate the robustness, stability, and transferability of the learned cluster structures.

12. Test on External Dataset (Fan-Made Pokémon)

A first generalization test was performed on a **fan-made dataset** containing Pokémon with the same numerical features as the official dataset.

12.1 Method

- The Stats and Power/Rarity features were extracted.
- The same scalers used during training were applied.
- The K-Means models ($k = 3$ for Stats, $k = 4$ for Power/Rarity) were used to assign clusters.
- Distances to centroids were computed to identify Pokémon that deviate strongly from the learned structure.

	Name	OG 'Mon	cluster	min_distance	hp	attack	defense	sp_attack	sp_defense	speed
333	Chansea	Chansey	1	7.014888	250.0	15.0	60.0	35.0	60.0	40.0
101	Sheeple	Shuckle	1	6.956111	255.0	50.0	105.0	30.0	60.0	5.0
334	Blissea	Blissey	1	6.878233	255.0	20.0	80.0	60.0	80.0	45.0
383	MissingNo	Original	0	5.645222	190.0	110.0	65.0	35.0	35.0	165.0
405	Regideep	Registeel	0	4.741342	80.0	100.0	50.0	50.0	200.0	100.0
261	Wobboogie	Wobuffet	1	4.615106	195.0	43.0	78.0	43.0	78.0	43.0
94	Ujadalord	Wailord	0	4.238005	35.0	135.0	45.0	95.0	45.0	185.0
109	Cloyseraph	Cloyster	0	4.151727	50.0	45.0	85.0	95.0	180.0	70.0
403	Registatic	Regirock	0	4.046437	80.0	200.0	50.0	100.0	50.0	100.0
93	Ujamer	Wailmer	0	3.864632	1.0	95.0	45.0	85.0	45.0	145.0
56	Piderpuff	Jigglypuff	2	3.707139	35.0	65.0	25.0	60.0	140.0	25.0
224	Skelegron	Aggron	0	3.700536	75.0	115.0	60.0	55.0	60.0	185.0
361	Kelvipass	Probopass	1	3.628486	60.0	55.0	150.0	85.0	150.0	40.0
436	Intriobang	Wugtrio	0	3.616513	65.0	50.0	65.0	170.0	145.0	105.0
426	Snorshoot	Original	1	3.616203	135.0	65.0	110.0	145.0	110.0	35.0
78	Pampaking	Slaking	0	3.579383	160.0	130.0	100.0	115.0	65.0	100.0

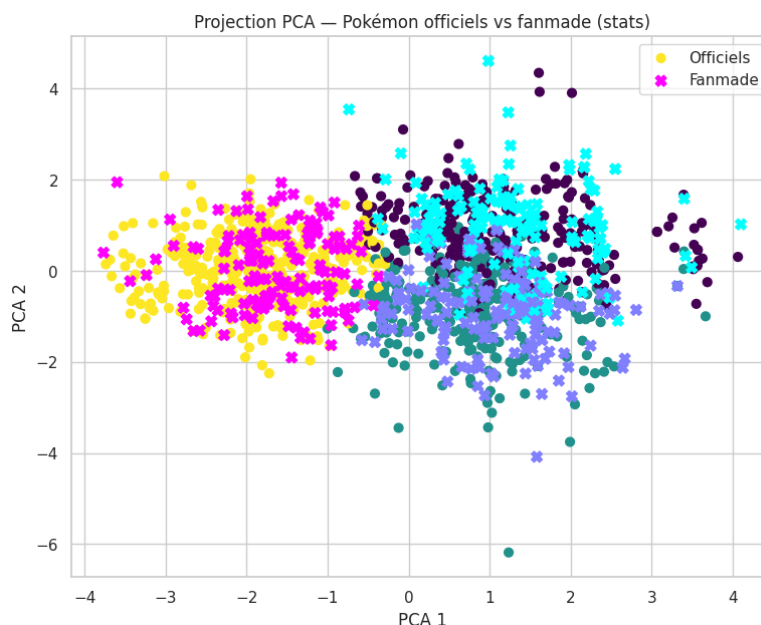
12.2 Interpretation of Outliers

The Pokémon with the largest distances to the centroids fall into three categories:

1. Pokémon whose stats were heavily modified. Their profiles diverge strongly from official distributions (e.g., extreme HP or Defense values). We have for example (e.g. screenshot): Wailmer, Shuckle...
2. Legendary or mythical Pokémon. Their stat spreads are unusually high or highly specialized, making them structurally distant from standard clusters. We have, for example (e.g. screenshot): Regirok, Registeel, Rayquaza...
3. Original fan-made Pokémon. Their design does not follow the statistical patterns of official generations, so they naturally fall outside the learned structure. We have for example (e.g. screenshot): MissingNo, Snorshot...

12.3 PCA Interpretation — Official Pokémon vs Fan-Made Pokémon (Stats)

The PCA projection shows that the fan-made Pokémon fall into **the same clusters** as the official Pokémon from previous generations. Their points overlap with the existing cluster structure, indicating that the K-Means model trained on Generations 1–8 generalizes well to external, non-official data. The fan-made Pokémon do not form new groups or isolated regions; instead, they integrate naturally into the established combat archetypes. This demonstrates that the clustering model is **robust, stable**, and capable of capturing structural patterns that remain valid even for artificially designed Pokémon.



12.4 Conclusion

The K-Means clusters generalize well: Pokémon that deviate significantly are precisely those expected to be structurally unusual.

13. Test on Official Generation 9 Pokémon

A second generalization test was performed on the official Generation 9 Pokémon, which was not included in the training dataset.

13.1 Method

- Gen 9 Pokémon were extracted using their Pokédex numbers.
- Stats and Power/Rarity features were selected.
- The same scalers and K-Means models were applied.
- Distances to centroids were computed to identify the most surprising Pokémon.

	name	cluster	distance_to_centroid		name	cluster	distance_to_centroid
974	cetitan	1	3.665155	986	flutter-mane	3	3.009975
991	iron-hands	1	3.092290	1023	terapagos	1	2.764634
986	flutter-mane	0	3.074887	1007	miraidon	0	2.682756
1002	ting-lu	1	2.857349	1006	koraidon	0	2.679896
979	clodsire	1	2.774373	1002	ting-lu	3	2.523814

13.2 Results

For both feature groups (Stats and Power/Rarity), in the five Pokémon farthest from the centroids, three or four are consistently Paradox Pokémon (which are a special type of Pokemons).

These Pokémon have semi-legendary profiles, with extreme or highly unusual stat distributions. Their design intentionally breaks the patterns of previous generations, which explains why they fall far from the learned clusters.

13.3 Interpretation

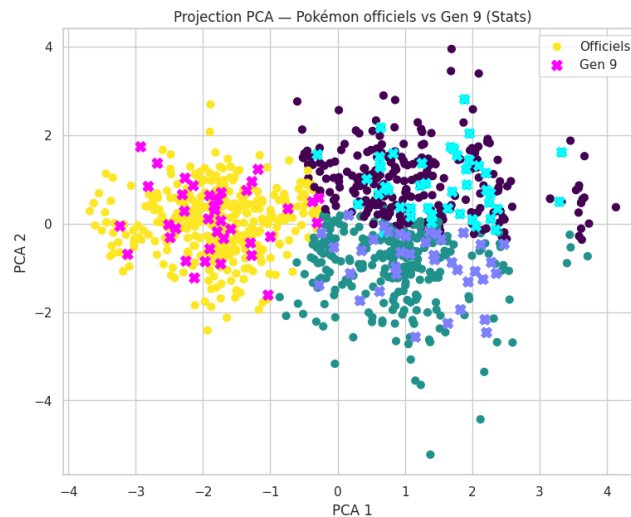
This result is coherent with the game design:

- Paradox Pokémon have very high base stats,
- unusual offensive/defensive spreads,
- and rarity attributes like legendary Pokémon.

Therefore, their distance from the centroids is expected and confirms that the model correctly identifies structural anomalies.

13.4 PCA Interpretation — Official Pokémon vs Generation 9 (Stats)

The PCA projection shows that the Generation 9 Pokémon falls into the same clusters as the official Pokémon from previous generations. Their points overlap with the existing cluster structure, indicating that the K-Means model trained on Generations 1–8 generalizes well to new data. Generation 9 Pokémon does not form new groups or isolated regions; instead, they integrate naturally into the established combat archetypes. This confirms the stability, consistency, and transferability of the clustering model.



14. Conclusion on K-Means Generalizability

The generalization tests demonstrate that:

- The clusters learned from Generations 1–8 are stable and transferable.
- Standard Pokémon from Gen 9 integrate naturally into the existing cluster structure.
- Only Pokémon with extreme, legendary, or non-standard profiles fall far from the centroids.
- This behavior is consistent across both feature families (Stats and Power/Rarity).

K-Means successfully captures structural patterns that remain valid across generations, and deviations from these patterns correspond to meaningful game-design exceptions (Paradox, legendary, or fan-made Pokémon)

PHASE 5 — Final Conclusion

The objective of this project was to determine whether Pokémon can be grouped into meaningful categories using unsupervised learning techniques, and to evaluate the stability and generalizability of these clusters across different feature families and datasets.

Through a structured methodology combining EDA, feature engineering, scaling, and clusterability analysis, we identified two complementary feature groups - Stats and Power/Rarity - each capturing a different dimension of Pokémon design. The clusterability study (Elbow Method, Silhouette Score, k-distance plots) revealed that Stats form a continuous structure with a clear separation at $k = 2-3$, while Power/Rarity exhibits a stronger natural organization around $k = 3-4$.

Applying K-Means, DBSCAN, and Agglomerative Clustering confirmed the robustness of these structures. K-Means provided the most interpretable segmentation, while DBSCAN and Agglomerative validated that the clusters correspond to real patterns rather than algorithmic artifacts. The centroid-ranking analysis allowed us to associate each cluster with coherent Pokémon categories, such as *fast special attackers*, *tanks*, *physical attackers*, *low-stat Pokémon*, *rare/powerful Pokémon*, and *common low-tier species*.

A key objective of the project was to evaluate whether these clusters generalize beyond the training data. Using the learned K-Means centroids, we assigned clusters to both a fan-made dataset and the official Generation 9 Pokémon. The results demonstrate strong generalizability: standard Pokémon integrate naturally into the learned structure, while outliers correspond to meaningful exceptions: legendary Pokémon, heavily modified fan-made species, and Paradox Pokémon with semi-legendary profiles.

Overall, the project successfully achieved its goals:

- identifying meaningful Pokémon categories through unsupervised learning,
- validating cluster stability across multiple algorithms,
- interpreting clusters through centroid ranking,
- and demonstrate that the learned structures generalize to new datasets.

This confirms that unsupervised learning provides a powerful and interpretable framework for analyzing Pokémon characteristics, revealing consistent patterns across generations and feature families.

Screenshots & Google Colab Environment

To ensure transparency and reproducibility of the analysis, this section provides screenshots of the main results as they appeared in the Google Colab environment. These include:

- the Elbow Method plots used to evaluate cluster compactness,
- the Silhouette Score curves used to assess cluster separation,
- the k-distance graph used for DBSCAN density estimation,
- and the PCA visualizations showing cluster structures and generalization to external datasets.

-

All computations were performed in Google Colab, using Python and the following key libraries: pandas, numpy, scikit-learn, matplotlib, and seaborn.

Screenshots of the main results as obtained in the Google Colab environment are provided below.

- 0. Implementation

```
##Elbow Method
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

def plot_elbow(data, max_k=12):
    inertias = []
    K = range(2, max_k+1)

    for k in K:
        kmeans = KMeans(n_clusters=k, random_state=42)
        kmeans.fit(data)
        inertias.append(kmeans.inertia_)

    plt.figure(figsize=(8,5))
    plt.plot(K, inertias, marker='o')
    plt.xlabel("Number of clusters (k)")
    plt.ylabel("Inertia")
    plt.title("Elbow Method")
    plt.grid(True)
    plt.show()
```

```
##Silhouette Score
from sklearn.metrics import silhouette_score

def compute_silhouette(data, max_k=12):
    scores = []
    K = range(2, max_k+1)

    for k in K:
        kmeans = KMeans(n_clusters=k, random_state=42)
        labels = kmeans.fit_predict(data)
        score = silhouette_score(data, labels)
        scores.append(score)

    plt.figure(figsize=(8,5))
    plt.plot(K, scores, marker='o')
    plt.xlabel("Number of clusters (k)")
    plt.ylabel("Silhouette Score")
    plt.title("Silhouette Score by Number of Clusters")
    plt.grid(True)
    plt.show()
```

... Afficher la sortie masquée

```
##K-distance plot
from sklearn.neighbors import NearestNeighbors
import numpy as np
import matplotlib.pyplot as plt

def plot_kdist(data_scaled, k=5):
    neigh = NearestNeighbors(n_neighbors=k)
    neigh.fit(data_scaled)
    distances, indices = neigh.kneighbors(data_scaled)
    distances = np.sort(distances[:, k-1])

    plt.figure(figsize=(8,6))
    plt.plot(distances)
    plt.title("k-distance graph")
    plt.xlabel("Sorted points")
    plt.ylabel(f"Distance for {k}-th neighbor")
    plt.grid(True)
    plt.show()
```

... k-distance graph

```

#Comparison with PCA
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
import numpy as np

official_scaled = df_stats_scaled
gen9_scaled = scaler_stats.transform(df_gen9[stats])

#PCA combination
combined = np.vstack([official_scaled, gen9_scaled])
pca = PCA(n_components=2)
combined_pca = pca.fit_transform(combined)

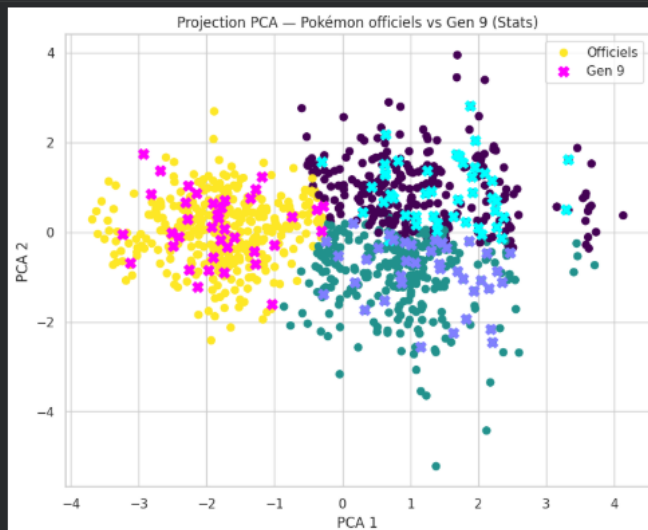
n_official = official_scaled.shape[0]
official_pca = combined_pca[:n_official]
gen9_pca = combined_pca[n_official:]

official_labels = k_means_stats_3.labels_
gen9_labels = k_means_stats_3.predict(gen9_scaled)

plt.figure(figsize=(9,7))
plt.scatter(
    official_pca[:,0], official_pca[:,1],
    c=official_labels, cmap="viridis", s=40, label="Officiels"
)
plt.scatter(
    gen9_pca[:,0], gen9_pca[:,1],
    c=gen9_labels, cmap="cool", s=80, marker="x", label="Gen 9"
)

plt.title("Projection PCA — Pokémon officiels (Gen 1-8) vs Gen 9 (Stats)")
plt.xlabel("PCA 1")
plt.ylabel("PCA 2")
plt.legend()
plt.grid(True)
plt.show()

```



Test on rarity/power

```
##Selection + Scaling
powers_col = ["base_stat_total", "capture_rate", "hatch_counter",
              "physical_total", "special_total", "offensive_total", "defensive_total" ]
gen9_power = df_gen9[powers_col]
gen9_power_scaled = scaler_power.transform(gen9_power)

##Prediction/Comparison
gen9_power_labels = k_means_power_4.predict(gen9_power_scaled)
gen9_power_distances = k_means_power_4.transform(gen9_power_scaled)

gen9_power_results = pd.DataFrame({
    "name": df_gen9["name"],
    "cluster": gen9_power_labels,
    "distance_to_centroid": gen9_power_distances.min(axis=1)
})

gen9_power_results.sort_values("distance_to_centroid", ascending=False).head(5)
```

	name	cluster	distance_to_centroid
986	flutter-mane	3	3.009975
1023	terapagos	1	2.764634
1007	miraidon	0	2.682756
1006	koraidon	0	2.679896
1002	ting-lu	3	2.523814

Results

For both the Stats and Power/Rarity feature groups, the five most surprising Generation 9 Pokémon — those farthest from the learned centroids — are consistently Paradox Pokémon, whose semi-legendary profiles naturally place them outside the standard cluster structure.

Group Contribution Summary

This project was primarily developed by Léo Carlier-Barnoux, who carried out the full technical workflow, including dataset exploration, preprocessing, feature engineering, clustering implementation, evaluation, visualization, and interpretation. All coding, methodological decisions, and analytical sections of the report were produced by Léo.

The other group members contributed by reviewing, formatting, and structuring the final report document. Their role focused on improving the presentation quality, ensuring consistency in layout, and helping organize the written material into a clear and readable format. No modifications were made to the analytical content, methodology, or results.

The final submission therefore reflects a collaborative effort in terms of document preparation, while the full technical and analytical work was completed by Léo Carlier-Barnoux.